

A FINITE-POOL AUDIT OF MULTIPLE-CHOICE SCORING PROTOCOLS FOR LLaDA-8B ON GSM8K-MC

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ABSTRACT

Multiple-choice accuracy for a diffusion language model depends on the scoring protocol as well as the model. We study this dependence on one bounded bed: LLaDA-8B-Instruct with constructed four-choice GSM8K-MC pools. The sole primary endpoint is the source-manuscript-designated preregistered comparison that assigns the random-mask reference L_i draws per candidate. On the $n=400$ pool, the source manuscript reports that s_conf exceeds $mc_uniform$ by 13.0 pp ($b/c = 106/54$, exact $p = 4.8 \times 10^{-5}$). A separately identified, post-selected $2L_i/2L_i$ forward-call sensitivity reports the same 13.0 pp gap, and a high-budget robustness endpoint reports 9.0 pp. On a disjoint $n=819$ pool, fixed left-to-right reveal is reported to exceed s_conf by 5.62 pp on average, while the direction reverses at answer position 0. The estimands are finite-pool contrasts conditional on the recorded stochastic scorer realizations. All empirical numbers are source-reported and cannot be independently verified from the current package; no experiment was rerun for this revision. The evidence therefore supports only a bed-local difference among complete scoring protocols, not an individual mechanism, stochastic-policy expectation, actual-cost comparison, or cross-bed law.

Evidence notation. Throughout, SR/CV means *source-reported; cannot be independently verified from the current package*. Every empirical numerical measurement in the abstract, prose, tables, and appendix has SR/CV status. Algebraic identities and declared design constants are not empirical measurements. The labels *preregistered* and *pre-specified* likewise describe the source manuscript; the registration objects needed to audit their timing are absent.

1 INTRODUCTION

A masked diffusion language model does not by itself determine a multiple-choice accuracy. A scorer must decide which candidate tokens remain masked, in what order tokens are revealed, whether known candidate tokens are fed back, how many model calls are used, and how token losses become one option score. Changing that tuple changes the measurement instrument.

We audit five such tuples on one LLaDA-8B-Instruct $\times$ GSM8K-MC bed. The contribution is not a new scorer or a component-level causal result. It is a bounded comparison that (i) separates nominal draw matching from a forward-call-count sensitivity, (ii) conditions stochastic comparisons on the recorded realizations, and (iii) exposes a fixed-order result whose aggregate direction does not hold at answer position 0. Table 1 is the canonical hierarchy for the entire paper.

Table 1: Canonical endpoint hierarchy. All effects are SR/CV. Positive signs follow the contrast printed in column two. Matched- L is the sole primary endpoint; no other row is a co-primary or replication.

Role	Explicit contrast	Pool / evidence ID	Reported effect	Interpretation boundary
Sole primary	$s_conf - mc_uniform, n_{mc,i} = L_i$	$n = 400$; E11	+13.0 pp	Nominal scoring-draw match only
Post-selected sensitivity	$s_conf - mc_uniform, 2L_i/2L_i$ calls	$n = 400$; E12	+13.0 pp	Forward-call-count sensitivity at one tested point
High-budget robustness	$s_conf - mc_uniform, n_{mc} = 128$	$n = 400$; E12	+9.0 pp	Smaller reported gap at one higher budget
Secondary, one schedule	$s_conf - rand_inject$	fresh $n = 819$; E13	+12.58 pp	Conditional on one recorded random schedule
Secondary, fixed order	$fixed_inject - s_conf$	fresh $n = 819$; E14	+5.62 pp	Aggregate reverses at position 0

The hierarchy prevents two common overreads. First, matching a reported count of forward calls is not a second confirmatory endpoint. Second, neither the random-order nor fixed-order comparison identifies a general benefit of a reveal policy. Both are conditional finite-pool observations under the particular choice constructions described below.

2 CLOSEST WORK AND CONTRIBUTION BOUNDARY

LLaDA introduced the masked-diffusion model family used here (Nie et al., 2025), and masked diffusion language modeling provides the broader objective and sampling context (Sahoo et al., 2024). The closest verified scorer source is iLLaDA (Nie et al., 2026). It defines a deterministic confidence-based multiple-choice surrogate that repeatedly reveals the ground-truth candidate token with greatest model confidence and sums its log probabilities. The paper explicitly treats that score as a task-specific surrogate rather than a likelihood estimate and compares it with a likelihood-style baseline on PIQA, ARC-Challenge, and HellaSwag. It does not report the present paired same-item audit, nominal-draw versus forward-call accounting, random-mask budget curve, reveal-order comparison, or GSM8K-MC construction analysis. Our contribution is therefore an audit of the measurement protocol, not the invention of confidence-guided scoring.

Prior work also shows that answer position can affect multiple-choice selection (Zheng et al., 2024). That result motivates caution around our position reversal but does not study masked-diffusion ground-truth-token injection or its call accounting. The current citation lock verifies these closest-work boundaries; a bibliography-wide metadata and sentence-support audit remains incomplete.

3 STUDY DEFINITION

Bed and evidence boundary. The source manuscript describes one frozen LLaDA-8B-Instruct checkpoint and four-choice pools derived from GSM8K (Cobbe et al., 2021). It reports two historical balanced pilots, an exactly answer-position-balanced $n = 400$ pool, and a disjoint fresh $n = 819$ pool with approximate position balance. The fresh pool mixes 336 near-miss items and 483 items whose distractors were drawn from a fixed answer pool. Arms are paired within each named pool and use the same recorded choice set. The current package contains neither item identifiers nor construction code, so pool membership, disjointness, prompts, and choice construction cannot be independently checked.

The source describes BF16 inference without generation or temperature. Confidence and fixed schedules are deterministic. The random-order result uses one stored schedule realization; the random-mask reference uses one stored collection of mask draws at each reported budget. The underlying orders and draws are not in the current package. Method statements in this section are therefore manuscript-described rather than code-verified.

Five scoring arms. For candidate tokens $y_{1:L_i}$, an injected arm reads a token loss, reveals the known candidate token at a selected position, feeds that token into later steps, and sums the per-step losses. `s_conf` selects the minimum-loss position, `anti_inject` the maximum-loss position, `rand_inject` a seeded-random order, and `fixed_inject` a fixed left-to-right order. The non-injected `mc_uniform` arm samples a masked subset of size $k \sim U\{1, \dots, L_i\}$ on each draw, rescales the partial-mask loss by L_i/k , and averages across draws. All arms select the option with minimum reported score, with first-index tie breaking.

Table 2: Manuscript-described arm decomposition. The implementation and call traces are absent, so the accounting cannot be independently verified.

Arm	Reveal schedule	GT feedback	Reported calls per candidate
<code>s_conf</code>	minimum loss first	yes	L_i score + L_i select
<code>rand_inject</code>	recorded random order	yes	L_i
<code>anti_inject</code>	maximum loss first	yes	L_i score + L_i select
<code>fixed_inject</code>	fixed left-to-right	yes	L_i
<code>mc_uniform</code>	random masked subsets	no	n_{mc}

The paper uses a *forward call* only as the source manuscript’s model- invocation accounting unit. Equality in that count does not establish equality of FLOPs, latency, memory, throughput, or energy. This is the only resource-accounting claim made in the paper; no broader equivalence is implied.

Finite-pool, realized-schedule estimand. Let $\mathcal{P}$ be one named finite pool and let ω denote the complete recorded realization of every stochastic scoring choice: random-mask subsets for `mc_uniform` and reveal orders for `rand_inject`. For deterministic arms, ω is empty. The target is

$$\Delta_{\mathcal{P},\omega}(A, B) = \frac{1}{|\mathcal{P}|} \sum_{q \in \mathcal{P}} [\mathbb{K}\{A_\omega(q) = y_q\} - \mathbb{K}\{B_\omega(q) = y_q\}]. \quad (1)$$

It is a descriptive contrast for the realized item set and scorer traces, not an expectation over a source population, mask streams, or random schedules. The source manuscript reports exact two-sided McNemar tests and intervals from paired item records. We retain those summaries as SR/CV but do not treat them as seed-aware uncertainty or as population coverage guarantees.

For paired correctness, the source defines b and c as the two discordant directions and reports $\hat{\Delta} = (b - c)/n$. It also reports a ten-test Holm family for the five-arm $n = 400$ analysis. The registration object and complete hypothesis ledger are absent, so family membership, adjusted values, and timing cannot be audited here. Table 1 retains the source manuscript’s role labels while making this evidence boundary explicit.

4 SOLE PRIMARY ENDPOINT AND BUDGET SENSITIVITIES

Table 3 reports the sole primary result and three budget comparators on the same named $n = 400$ pool. E11 designates the missing matched- L record; E12 designates the missing $2L$ and 128-draw records; E10 designates the missing single-draw record. These are opaque paper-local IDs, not resolvable artifacts in the current package, and no crosswalk is available.

Table 3: SR/CV reference-budget contrasts on the reported $n = 400$ pool. Intervals and exact two-sided McNemar tests are transcribed from the source manuscript. The role column is the canonical hierarchy; only matched- L is primary.

Reference budget (ref./s_conf reported calls)	n	Acc. pair (s_conf/ref.)	b/c	$\hat{\Delta}$	95% interval	Exact p	Role
Single draw (1/2L)	400	42.75/28.75%	113/57	+14.0 pp	[7.61, 20.39]	2.1×10^{-5}	corroborative curve point (E10)
$n_{mc,i} = L_i$ (L/2L)	400	42.75/29.75%	106/54	+13.0 pp	[7.00, 19.00]	4.8×10^{-5}	sole primary (E11)
$n_{mc,i} = 2L_i$ (2L/2L)	400	42.75/29.75%	109/57	+13.0 pp	[6.75, 19.25]	6.6×10^{-5}	post-selected sensitivity (E12)
$n_{mc} = 128$ (128/2L)	400	42.75/33.75%	82/46	+9.0 pp	[3.46, 14.54]	0.0019	high-budget robustness (E12)

At the sole primary point, both arms receive L_i nominal scoring draws, but the manuscript ledger assigns s_conf another L_i schedule-selection calls. The source-reported finite-pool gap is +13.0 pp. The post-selected 2L/2L row has the same reported gap but a different discordance record. Conditional on the manuscript’s unverified call ledger, this sensitivity addresses only the explanation that the reference is lower because it receives fewer counted forward calls at that tested point. It neither becomes a second primary result nor isolates feedback, reveal order, or estimator choice.

The 128-draw robustness row has the smallest reported gap on the displayed curve. The reference changes with budget, so the four rows are operating points of one protocol replacement rather than independent replications. Nothing here supports a dose-response form or behavior between or beyond the tested budgets.

5 SECONDARY REVEAL-ORDER RESULTS

Reveal-order evidence addresses two different questions and is reported in two directions. Table 4 uses s_conf minus rand_inject and is conditional on one recorded random schedule. Table 5 uses fixed_inject minus s_conf throughout, so a positive value always favors fixed left-to-right reveal.

5.1 ONE RECORDED RANDOM SCHEDULE

Table 4: SR/CV contrasts for s_conf minus rand_inject. These rows concern one recorded schedule realization, not the expectation over random schedules. The pooled row combines heterogeneous pools and is only a disclosure.

Pool	n	$\hat{\Delta}$	m	Role
Fresh	819	+12.58 pp	305	post-selection secondary (E13)
$n = 400$ pilot	400	+3.75 pp	141	undetermined
Pooled disclosure	1219	+9.68 pp	446	heterogeneous; not a common effect

The source manuscript reports that the fresh and pilot magnitudes differ ($z = 2.41$, two-sided $p = 0.016$) even with the stored schedule fixed. That comparison is itself SR/CV and its analysis object is missing. We therefore do not average over schedules, treat the pooled value as a transferable constant, or infer that confidence ordering is generally better than random ordering. Multi-schedule evaluation remains an evidence blocker.

5.2 FIXED LEFT-TO-RIGHT REVEAL AND THE POSITION REVERSAL

Table 5: SR/CV contrasts for `fixed_inject` minus `s_conf`; positive values favor fixed order. The position-0 row is adjacent to the aggregate so its reversal is explicit. Tests are the source-reported exact two-sided McNemar values.

Scope	n	b/c	$\hat{\Delta}$	Exact p	Role
Fresh aggregate	819	145/99	+5.62 pp	0.0039	secondary (E14)
Fresh, answer position 0	199	21/46	−12.56 pp	0.0031	post-hoc reversal (E14)

The fresh aggregate favors fixed order, whereas answer position 0 favors `s_conf`. Construction also matters: the source reports fixed-order margins of 2.98 pp in the near-miss stratum and 7.45 pp in the answer-pool stratum (Appendix Table 6). The position analysis is post-hoc, and no tie-robust interaction analysis is available. These records reject a unique-confidence interpretation only for the realized fresh pool; they do not show that confidence ordering is generally unnecessary.

6 LIMITATIONS, AVAILABILITY, AND DISCLOSURE

Scientific scope. The study covers one checkpoint and constructed four-choice pools from one task. The primary swap changes feedback, masking path, estimator, and reported call count together. The random-order result conditions on one unavailable schedule trace, while the fixed-order result depends on construction and answer position. The evidence supports neither component-level causality nor an expectation over scorer randomness. Cross-model, cross-task, multi-seed, and independently constructed-pool behavior remain unmeasured here.

What the current package contains. The authoring directory inspected for this revision contains `main.tex`, `appendix_refine5.tex`, `refs.bib`, the ICLR style files, a compiled manuscript PDF, and pre-rendered figure files. This revision does not include any figure in the manuscript. The directory does not contain the item-level experimental records, recorded schedules, machine-readable registrations, protocol or analysis code, figure-generation scripts, environment lock, or an ID-to-file crosswalk required to verify E10–E15. Thus the empirical values were neither rerun nor independently recomputed for this revision.

AI assistance disclosure. AI tools assisted with language editing, $\LaTeX$ refactoring, and isolated internal pre-submission review for this revision. The authors remain responsible for the manuscript’s claims and evidence boundaries. AI assistance did not provide experimental records, generate new measurements, or verify the source-reported results.

7 CONCLUSION

The sole primary endpoint is the matched- L `s_conf`–`mc_uniform` contrast on the named $n = 400$ pool, for which the source manuscript reports +13.0 pp. The $2L/2L$ result is a post-selected forward-call sensitivity, and the 128-draw result is a high-budget robustness endpoint. Separately, fixed order is reported to win on average on the fresh pool while losing at answer position 0. Interpreted as finite-pool contrasts conditional on the recorded scorer realizations, these observations support bed-local protocol non-interchangeability. Missing records, registrations, implementations, and multi-seed evidence prevent stronger empirical or causal conclusions.

REFERENCES

- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021. Venue: arXiv preprint; volume and pages not applicable.
- Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. Large language diffusion models, 2025. Venue: ICML 2025.

Shen Nie, Qiyang Min, Shaoxuan Xu, Zihao Huang, Yuxuan Song, Yong Shan, Yankai Lin, Wayne Xin Zhao, Chongxuan Li, and Ji-Rong Wen. Improved large language diffusion models, 2026. Venue: arXiv preprint; volume and pages not applicable.

Subham Sekhar Sahoo, Marianna Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin Chiu, Alexander Rush, and Volodymyr Kuleshov. Simple and effective masked diffusion language models. In *Proceedings of NeurIPS*, 2024.

Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust multiple choice selectors. In *Proceedings of ICLR*, 2024.

A EVIDENCE BOUNDARY AND POOL DESCRIPTION

Status of every measurement. As stated in the main text, every empirical numerical value in this appendix is SR/CV: source-reported and not independently verifiable from the current package. No missing record was reconstructed, no experiment was rerun, and no new measurement was generated during revision. The tables retain selected source values only to make the paper’s conditional claims and evidence gaps auditable at the prose level.

Finite pools and pairing. The source manuscript describes an exactly answer-position-balanced $n = 400$ four-choice pool and a disjoint fresh $n = 819$ four-choice pool with approximate position balance. All arms reportedly score the same recorded choice set within a pool, so the accuracy contrasts are paired by question. The first 336 fresh items use near-miss distractors; the remaining 483 use three distinct values drawn from a fixed answer pool. Neither the item manifest nor the distractor-construction implementation is present. The finite-pool estimand in Equation 1 therefore conditions on the named pools exactly as reported and does not assert a sampling mechanism or population target.

Stochastic scorer realizations. The source says that `rand_inject` uses one persisted schedule and that `mc_uniform` uses recorded mask draws at each budget. It also reports that the original random schedule depended on a process-sensitive string hash, making the nominal seed insufficient for replay. The per-candidate schedules and mask draws are absent from the package. Consequently, the random-order and random-mask results are conditional on unavailable realized traces, and a schedule-family or mask-stream expectation is unidentified.

B FIXED-ORDER CONSTRUCTION AND POSITION DETAIL

Table 6 uses one direction throughout: `fixed_inject` minus `s_conf`. Positive values favor fixed left-to-right reveal; negative values favor confidence ordering. This orientation is the same as Table 5.

Table 6: SR/CV fresh-pool fixed-order contrasts, consistently oriented as `fixed_inject` minus `s_conf`. The construction split and answer-position split are post-hoc re-presentations of the source manuscript’s aggregate record. The position rows do not constitute a formal interaction test.

Scope	n	b/c	$\hat{\Delta}$	Exact p	Reading
Near-miss construction	336	61/51	+2.98 pp	0.395	fixed direction; not detected
Answer-pool construction	483	84/48	+7.45 pp	0.0022	fixed direction
Answer position 0	199	21/46	−12.56 pp	0.0031	reversal: <code>s_conf</code> direction
Answer position 1	207	37/15	+10.63 pp	0.0032	fixed direction
Answer position 2	226	44/18	+11.50 pp	0.0013	fixed direction
Answer position 3	187	43/20	+12.30 pp	0.0052	fixed direction

The source manuscript reports a Bonferroni–4 comparison of the four within-position tests. That calculation and the score vectors needed for tie-robust or construction-adjusted interaction analyses are not present. Accordingly, the main text reports the sign reversal descriptively and does not claim a general answer-position mechanism.

C REPORTED BUDGET AND LENGTH ACCOUNTING

Protocol tuple. The complete interventions differ along several axes. The matched- L primary changes feedback, masking path, estimator, and the reported number of selection calls while matching nominal scoring draws. The $2L/2L$ post-selected sensitivity retains those protocol changes and matches only the source manuscript’s count of model invocations. The fixed-order and random-order comparisons retain ground-truth feedback but change reveal order; confidence ordering also carries reported schedule-selection calls. None of these comparisons isolates a single component.

Table 7: Manuscript-described per-candidate accounting. These are protocol definitions and reported call counts, not measured resource costs. The implementation and instrumented call traces are absent.

Protocol	Scoring draws	Selection calls	Total calls	Role
s_conf / anti_inject	L	L	$2L$	injected, model-derived order
rand_inject / fixed_inject	L	0	L	injected, recorded random or fixed order
mc_uniform single draw	1	0	1	corroborative budget point
mc_uniform matched- L	L	0	L	sole primary reference budget
mc_uniform matched- $2L$	$2L$	0	$2L$	post-selected forward-call sensitivity
mc_uniform high budget	128	0	128	robustness endpoint

Table 8: SR/CV candidate-length counts transcribed from the source manuscript. The records needed to verify tokenization and these distributions are absent.

Pool	Items	$L = 3$	$L = 4$	$L = 5$	$L = 6$	$L = 7$	$L = 8$	$L = 9$	Mean L
Historical	64	39	120	72	17	8	0	0	4.355
$n = 400$ reference pool	400	262	751	424	118	30	15	0	4.343
Fresh pool	819	364	1237	1010	390	167	101	7	4.722

Uncertainty and multiplicity. The source manuscript reports question-bootstrap intervals for matched- L and $2L/2L$, and a paired standard-error interval for the single-draw and 128-draw rows. It reports exact two-sided McNemar tests and a ten-test Holm family for the five-arm $n = 400$ analysis. The underlying item records, bootstrap draws, full hypothesis list, adjusted values, statistical code, and registration are unavailable. Table 3 therefore preserves the reported intervals and raw exact tests as SR/CV without claiming local reproduction or auditable familywise control.

D OPAQUE EVIDENCE-ID MANIFEST

The raw internal paths, run timestamps, trial identifier, and commit prefix in the earlier draft have been removed. Table 9 uses only opaque paper-local IDs. No ID resolves to a file in the current package, and the confidential crosswalk was not available for this revision.

Table 9: Required evidence objects. Quantitative scope values are SR/CV; every object and the ID crosswalk are absent from the current package.

ID	Intended object	Current status
E10	Single-draw paired record, reported $n = 400$	missing
E11	Matched- L sole-primary paired record, reported $n = 400$	missing
E12	$2L/2L$ sensitivity and 128-draw robustness records, reported $n = 400$	missing
E13	Fresh s_conf/rand_inject record and stored schedule, reported $n = 819$	missing
E14	Fresh s_conf/fixed_inject record with construction, position, and tie fields, reported $n = 819$	missing
E15	Reference chance-calibration ledger	missing

Table 10: Minimum contents of a complete anonymous experimental bundle. This is a requirements checklist, not evidence that the components are present.

Component	Required contents	Independent check
Sampling frames	Item IDs, split membership, exclusions, pool assignment, and overlap report	Verify pool sizes, balance, and disjointness
Choice construction	Distractor rules, deduplication, shuffling, answer positions, and candidate strings	Rebuild and byte-compare every choice set
Prompts and model	Prompt bytes; model and tokenizer revisions; precision and environment	Reconstruct tokenization and scoring inputs
Protocol traces	Implementations, configurations, tie policy, mask draws, and reveal orders	Replay each arm and verify score vectors and predictions
Statistics	Paired records, interval and exact-test code, complete hypothesis families, and registrations	Recompute every reported statistic and role label
Figures	Deterministic scripts and numeric payloads for any quantitative figure	Regenerate and compare plotted values
Manifest	Content hashes, environment lock, and canonical command sequence	Validate the bundle before recomputation

Exact package statement. The authoring directory used for this revision contains the two $\text{\LaTeX}$ sources, `refs.bib`, ICLR style files, the compiled PDF, and pre-rendered figure files. No figure is included by the revised $\text{\LaTeX}$ source. The directory does not contain any E10–E15 object or the experimental components in Table 10. This distinction is why the paper labels every empirical value SR/CV.

E CITATION-AUDIT BOUNDARY

The citation lock verifies the primary-source metadata and local support used for the closest-work boundaries of iLLaDA, LLaDA, the multiple-choice position-sensitivity study, and masked diffusion language modeling (Nie et al., 2026; 2025; Zheng et al., 2024; Sahoo et al., 2024). In particular, iLLaDA supplies the closest verified scoring mechanism but not this paper’s paired finite-pool protocol and budget audit. The lock does not claim a complete bibliography-wide audit. Remaining entries and broad background statements must be checked against primary sources before a full citation-verification claim can be made.